

DDR Memory Generations, DFI Protocol Versions & Memory Controller Interfaces

Comprehensive Technical Reference

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Covering: DDR1 through DDR5 · LPDDR1–LPDDR5X · GDDR5/6 · HBM1/2/3
DFI 2.1 – DFI 5.1 · AXI Memory Controller Interface · PHY Interface Signals

March 2026

1. Introduction & Background

Modern SoC memory subsystems comprise three primary layers: the **Memory Controller (MC)**, the **Physical Layer (PHY)**, and the **DRAM device**. Standardising the interface between the MC and the PHY is the goal of the **DFI (DDR PHY Interface)** specification, maintained by the DFI PHY Interface Working Group. On the SoC bus side, the MC exposes an **AXI** (Advanced eXtensible Interface) slave port to system interconnects and ARM interconnects expose an **AXI** master that issues read/write transactions to the MC.

DDR	Double Data Rate SDRAM	DFI	DDR PHY Interface
LPDDR	Low Power DDR	AXI	Advanced eXtensible Interface
GDDR	Graphics DDR	PHY	Physical Layer
HBM	High Bandwidth Memory	MC	Memory Controller
JEDEC	Joint Electron Device Eng. Council	CA	Command/Address bus
BL	Burst Length	tRCD	RAS-to-CAS Delay

The evolution of DDR standards has been driven by increasing bandwidth demands, lower supply voltages for energy efficiency, and finer-grained error correction. Correspondingly, DFI has evolved from a simple control/data split in DFI 2.1 to full LPDDR5X, HBM3, and CXL-attached DRAM support in DFI 5.1.

2. DDR SDRAM Generations — Full Comparison

2.1 Core Electrical & Architectural Parameters

Parameter	DDR1	DDR2	DDR3	DDR4	DDR5
JEDEC Std	JESD79	JESD79-2	JESD79-3	JESD79-4	JESD79-5
Year Introduced	2000	2003	2007	2012	2020
VDD (V)	2.5	1.8	1.5	1.2	1.1
VDDQ (V)	2.5	1.8	1.5	1.2	1.1
Prefetch	2n	4n	8n	8n	16n
Data Rate (MT/s)	200–400	400–1066	800–2133	1600–3200	3200–8400
Peak BW (GB/s)	3.2	8.5	17	25.6	67.2
Bus Width (bits)	64	64	64	64	64 (2×32)
Die Capacity	64Mb–1Gb	256Mb–4Gb	512Mb–8Gb	2Gb–16Gb	8Gb–64Gb
Banks	4	4/8	8	16	32
Bank Groups	–	–	–	2 or 4	8
Burst Length	2/4	4/8	8	8/4(BC)	16/8(BC)
CAS Latency	2–3	3–7	5–11	9–19	22–52
Signalling	SSTL 2.5	SSTL 1.8	SSTL 1.5	POD 1.2	SCSI/POD 1.1
ECC (on-die)	× No	× No	× No	× No	✓ Yes
On-Die Term.	× No	✓ Yes	✓ Yes	✓ Yes	✓ Yes
Write CRC	× No	× No	× No	✓ Yes	✓ Yes
Package	TSOP	FBGA	FBGA	FBGA	FBGA
CA Parity	× No	× No	× No	✓ Yes	✓ Yes
Pseudo-Open Drain	× No	× No	× No	✓ Yes	✓ Yes
Decision Feedback Eq.	× No	× No	× No	Optional	✓ Yes
Sub-channel	× No	× No	× No	× No	✓ Yes(2×40b)
Power Mgmt IC	× No	× No	× No	× No	✓ Yes(onboard PMIC)

Table 1: DDR1–DDR5 Core Parameters

2.2 Timing & Latency Comparison

Timing	DDR1	DDR2	DDR3	DDR4	DDR5
tCK min (ns)	5.0	1.875	0.938	0.625	0.238
tRCD (ns)	15–20	12.5–15	10–12.5	12.5–15	14
tRP (ns)	15–20	12.5–15	10–12.5	12.5–15	14
tRAS (ns)	40–50	40–45	35–40	32–35	32
tRC (ns)	55–70	52.5–60	46–52	46.5–50	46
tREFI (μs)	7.8	7.8	7.8	7.8	3.9
tRFC (ns)	75	75–127	110–350	160–550	295–880
Read Latency	CL+AL	CL+AL	CL+AL	CL+AL	CL+AL
Write Latency	WL	WL	WL	WL	WL
tDQSS (tCK)	± 0.25	± 0.25	± 0.25	± 0.27	± 0.27

Table 2: DDR Timing Parameters by Generation

3. LPDDR Generations — Full Comparison

3.1 Low Power DDR Parameters

Parameter	LPDDR1	LPDDR2	LPDDR3	LPDDR4	LPDDR4X	LPDDR5/5
JEDEC Std	JESD209	JESD209-2	JESD209-3	JESD209-4	JESD209-4	JESD209-5
Year	2003	2009	2012	2014	2016	2019/2021
VDD (V)	1.8	1.2	1.25	1.1	0.6	0.5
VDDQ (V)	1.8	1.2	1.2	0.6	0.6	0.5
Prefetch	1n	4n	8n	16n	16n	16n
Data Rate (Gbps)	0.2–0.4	0.5–1.066	0.8–2.133	1.6–4.267	2.133–4.267	6.4–8.533
Peak BW/ch (GB/s)	3.2	8.5	17	34	34	68
Bus Width	16/32b	16/32b	16/32b	16/32b	16/32b	16/32b
CA Bus	10-bit	10-bit	10-bit	6-bit	6-bit	6-bit
Clock Type	SDR CK	Diff CK	Diff CK	Diff CK	Diff CK	Diff CK
On-Die Term	× No	✓ Yes	✓ Yes	✓ Yes	✓ Yes	✓ Yes
Write Leveling	× No	✓ Yes	✓ Yes	✓ Yes	✓ Yes	✓ Yes
Data Bus Inv.	× No	× No	✓ Yes	✓ Yes	✓ Yes	✓ Yes
Bank Arch	4 banks	8 banks	8 banks	8 BG×4	8 BG×4	8 BG×4
Self Ref. Ctrl	Basic	Basic	Temp-aware	Fine-grained	Fine-grained	Fine-grained
Refresh Mode	auto	auto	auto	per-bank	per-bank	per-bank
Link ECC	× No	× No	× No	× No	× No	✓ Yes
Write CRC	× No	× No	× No	✓ Yes	✓ Yes	✓ Yes
Deep Pwr Down	✓ Yes	✓ Yes	✓ Yes	✓ Yes	✓ Yes	✓ Yes
Target Use	Mobile	Mobile	Mobile	Mobile/IoT	Mobile	Mobile/AI

Table 3: LPDDR1 through LPDDR5X Parameters

4. GDDR and HBM Memory Generations

4.1 GDDR Comparison (Graphics DDR)

Parameter	GDDR3	GDDR4	GDDR5	GDDR5X	GDDR6/6X
JEDEC Std	JESD79-3G	JESD79-4G	JESD212	JESD212C	JESD250
Year	2004	2006	2008	2016	2018/2019
VDD (V)	1.8	1.8	1.5	1.35	1.25/1.35
Prefetch	4n	8n	8n	16n	16n
Data Rate (Gbps/pin)	1.6	3.2	8	11.2	16/22
Bus Width (bits)	32	32	32	32	32
Signalling	HSTL	HSTL	HSTL	HSTL	POD/PAM-4
PAM-4	× No	× No	× No	× No	GDDR6X only
QDR clocking	× No	× No	✓ Yes	✓ Yes	✓ Yes
Peak BW/device	12.8 GB/s	25.6	64	89.6	128/176
Target	GPU	GPU	GPU	GPU	GPU/AI

Table 4: GDDR Memory Generations

4.2 HBM Comparison (High Bandwidth Memory)

Parameter	HBM1	HBM2	HBM2E	HBM3
JEDEC Std	JESD235	JESD235B	JESD235C	JESD238
Year	2013	2016	2018	2022
VDD (V)	1.2	1.2	1.2	1.1
I/O bus width	128b per stack	128b per stack	128b per stack	128b per stack
Stacks per device	4 die	8 die	12 die	12–16 die
Total bus width	1024b	1024b	1024b	1024b
Data Rate (Gbps/pin)	1.0	2.0	3.6	6.4
Peak BW/s-stack	128 GB/s	256 GB/s	460 GB/s	819 GB/s
TSV pitch (μm)	55	55	55	55
ECC	✓Yes	✓Yes	✓Yes	✓Yes(SECDED)
Pseudo channel	× No	× No	× No	✓Yes(2×512b)
DBI	× No	✓Yes	✓Yes	✓Yes
Power (typ W)	35	25	25	30
Target	GPU HPC	GPU AI	HPC AI	AI/HPC

Table 5: HBM1 through HBM3 Parameters

5. DFI (DDR PHY Interface) — Version History & Features

The **DDR PHY Interface (DFI)** specification defines a standardised bus between a *Memory Controller (MC)* and a *PHY* in an SoC or ASIC. It abstracts the DRAM protocol complexity from the MC, allowing PHY vendors and MC vendors to develop interoperable IP blocks. DFI carries: command/address signals, data signals (DQ/DQS/DM), clocking, training control, low-power interface, and debug/status.

5.1 DFI Version Overview

Version	Year	Supported DRAM	Key Additions	Notable Features
DFI 1.0	2005	DDR1, DDR2	Initial standard	Basic ctrl/data split, single freq
DFI 2.0	2007	DDR2, DDR3	LPDDR support	Power mgmt, LP ctrl signals
DFI 2.1	2009	DDR3, LPDDR2	Training interface	Per-bit deskew, DFI training ctrl
DFI 3.0	2011	DDR3, LPDDR2	Enhanced training	dfi_rdlvl, dfi_wrlvl, gate training
DFI 3.1	2013	DDR4, LPDDR3	DDR4 support	Bank groups, CA training
DFI 4.0	2016	DDR4, LPDDR4	Full LPDDR4	Per-rank ODT, freq change
DFI 5.0	2020	DDR5, LPDDR5, HBM3	DDR5/LPDDR5	Sub-channel, ALERT_n, DRAM ECC
DFI 5.1	2022	DDR5, LPDDR5X, HBM3	LPDDR5X, CXL	RFM, PPR, fine-grained pwr

Table 6: DFI Version Summary

5.2 DFI Signal Group Comparison Across Versions

Feature / Signal Group	DFI 1.0	DFI 2.0	DFI 2.1	DFI 3.0	DFI 3.1	DFI 4.0	DFI 5.0	DFI 5.1
Command/Add	✓Yes	✓Yes	✓Yes	✓Yes	✓Yes	✓Yes	✓Yes	✓Yes
Data (DQ/DQS)	✓Yes	✓Yes	✓Yes	✓Yes	✓Yes	✓Yes	✓Yes	✓Yes
Write Leveling	× No	× No	✓Yes	✓Yes	✓Yes	✓Yes	✓Yes	✓Yes
Gate Training	× No	× No	× No	✓Yes	✓Yes	✓Yes	✓Yes	✓Yes
Read Leveling	× No	× No	✓Yes	✓Yes	✓Yes	✓Yes	✓Yes	✓Yes
LP Interface (CKE/SR)	× No	✓Yes	✓Yes	✓Yes	✓Yes	✓Yes	✓Yes	✓Yes
Freq Change (FC)	× No	× No	× No	✓Yes	✓Yes	✓Yes	✓Yes	✓Yes
Per-bit Deskew	× No	× No	✓Yes	✓Yes	✓Yes	✓Yes	✓Yes	✓Yes
CA Parity/-Training	× No	× No	× No	× No	✓Yes	✓Yes	✓Yes	✓Yes
VREF Training	× No	× No	× No	× No	× No	✓Yes	✓Yes	✓Yes
Sub-channel	× No	× No	× No	× No	× No	× No	✓Yes	✓Yes
DRAM ECC (ALERT)	× No	× No	× No	× No	× No	× No	✓Yes	✓Yes
RFM (Refresh Mgmt)	× No	× No	× No	× No	× No	× No	× No	✓Yes
PPR (Post Pkg Repair)	× No	× No	× No	× No	× No	× No	× No	✓Yes
Per-rank ODT	× No	× No	× No	× No	× No	✓Yes	✓Yes	✓Yes
HBM Support	× No	× No	× No	× No	× No	Partial	✓Yes	✓Yes
LPDDR5X	× No	× No	× No	× No	× No	× No	× No	✓Yes
Debug/Status IF	× No	× No	× No	✓Yes	✓Yes	✓Yes	✓Yes	✓Yes

Table 7: DFI Feature Matrix by Version

6. DFI Signal Groups — Detailed Description

6.1 Control & Command Signals (MC → PHY)

Signal Name	Direction	Description	Intro Ver.
dfi_address[n:0]	MC→PHY	Row/column/bank address bus	1.0
dfi_bank[n:0]	MC→PHY	Bank select bits	1.0
dfi_cas_n	MC→PHY	Column Address Strobe (active low)	1.0
dfi_ras_n	MC→PHY	Row Address Strobe (active low)	1.0
dfi_we_n	MC→PHY	Write Enable (active low)	1.0
dfi_cs_n[r:0]	MC→PHY	Chip Select per rank	1.0
dfi_cke[r:0]	MC→PHY	Clock Enable per rank	1.0
dfi_odt[r:0]	MC→PHY	On-Die Termination per rank	2.0
dfi_reset_n	MC→PHY	DRAM reset signal (DDR3+)	2.1
dfi_act_n	MC→PHY	Activate command (DDR4+)	3.1
dfi_bg[n:0]	MC→PHY	Bank Group select (DDR4+)	3.1
dfi_cas_n_p1	MC→PHY	Phase-1 CA (2T mode)	3.0
dfi_alert_n	PHY→MC	DRAM Alert (parity error)	5.0

Table 8: DFI Command/Address Signals

6.2 Data Path Signals

Signal Name	Direction	Description	Intro Ver.
dfi_wrdata[2n-1:0]	MC→PHY	Write data (2× bus width wide)	1.0
dfi_wrdata_en	MC→PHY	Write data enable strobe	1.0
dfi_wrdata_mask[n-1:0]	MC→PHY	Byte mask for write data	1.0
dfi_rddata[2n-1:0]	PHY→MC	Read data captured from DRAM	1.0
dfi_rddata_valid	PHY→MC	Read data valid indicator	1.0
dfi_rddata_en	MC→PHY	Read data enable (gate train)	3.0
dfi_wrdata_cs_n[r:0]	MC→PHY	CS for write operation (DDR4+)	3.1
dfi_rddata_cs_n[r:0]	MC→PHY	CS for read (DDR4+)	3.1
dfi_wrdata_p[1:0]	MC→PHY	Sub-channel write data (DDR5)	5.0
dfi_rddata_p[1:0]	PHY→MC	Sub-channel read data (DDR5)	5.0
dfi_wrdata_ecc	MC→PHY	ECC data (link ECC, LPDDR5)	5.0

Table 9: DFI Data Path Signals

6.3 Training Interface Signals

Signal Name	Direction	Description	Intro Ver.
dfi_wrlvl_en	MC→PHY	Enable write leveling mode	2.1
dfi_wrlvl_strobe	MC→PHY	Strobe signal for WL sampling	2.1
dfi_wrlvl_resp	PHY→MC	Write level response	2.1
dfi_rdlvl_en	MC→PHY	Enable read leveling	3.0
dfi_rdlvl_resp	PHY→MC	Read level response	3.0
dfi_rdlvl_gate_en	MC→PHY	Enable gate training	3.0
dfi_calvl_en	MC→PHY	CA leveling enable (LPDDR3+)	3.1
dfi_calvl_resp	PHY→MC	CA leveling response	3.1
dfi_vref_en	MC→PHY	VREF training enable	4.0
dfi_dram_clk_disable	MC→PHY	Disable DRAM CK output	2.0
dfi_phyupd_req	PHY→MC	PHY requests MC to pause	3.0
dfi_phyupd_ack	MC→PHY	MC acknowledges pause	3.0
dfi_ctrlupd_req	MC→PHY	MC requests ctrl update	3.0
dfi_ctrlupd_ack	PHY→MC	PHY acknowledges	3.0

Table 10: DFI Training Interface Signals

6.4 Low-Power Interface Signals

Signal Name	Direction	Description	Intro Ver.
dfi_lp_ctrl_req	MC→PHY	LP request to PHY	2.0
dfi_lp_ctrl_wakeup[3:0]	MC→PHY	Wakeup latency for LP ctrl	2.0
dfi_lp_ctrl_ack	PHY→MC	LP ctrl acknowledgement	2.0
dfi_lp_data_req	MC→PHY	LP request for data path	2.0
dfi_lp_data_wakeup[3:0]	MC→PHY	Wakeup latency for data LP	2.0
dfi_lp_data_ack	PHY→MC	Data LP acknowledgement	2.0
dfi_freq_ratio[1:0]	MC→PHY	Ratio of DFI clk to DRAM clk	3.0
dfi_freq_fsp[1:0]	MC→PHY	Freq Set Point for LPDDR4	4.0
dfi_init_complete	PHY→MC	PHY init done signal	2.1
dfi_init_start	MC→PHY	MC starts PHY init	2.1

Table 11: DFI Low-Power Interface Signals

7. DFI Version vs. DDR Generation Support Matrix

DFI Ver.	DDR1	DDR2	DDR3	DDR4	DDR5	LPDDR	LPDDR	LPDDR	LPDDR	HBM2/3
DFI 1.0	✓Yes	✓Yes	× No	× No	× No	× No	× No	× No	× No	× No
DFI 2.0	✓Yes	✓Yes	× No	× No	× No	Partial	× No	× No	× No	× No
DFI 2.1	✓Yes	✓Yes	✓Yes	× No	× No	✓Yes	× No	× No	× No	× No
DFI 3.0	✓Yes	✓Yes	✓Yes	× No	× No	✓Yes	Partial	× No	× No	× No
DFI 3.1	✓Yes	✓Yes	✓Yes	Partial	× No	✓Yes	✓Yes	× No	× No	× No
DFI 4.0	Leg	Leg	✓Yes	✓Yes	× No	✓Yes	✓Yes	✓Yes	× No	Partial
DFI 5.0	Leg	Leg	Leg	✓Yes	✓Yes	✓Yes	✓Yes	✓Yes	✓Yes	✓Yes
DFI 5.1	Leg	Leg	Leg	✓Yes	✓Yes	✓Yes	✓Yes	✓Yes	✓Yes	✓Yes

Table 12: DFI Version vs. DRAM Type Support (✓ = Fully Supported, Partial = Partial Support, Leg = Legacy/Deprecated)

- **Sub-channel support:** DDR5 and HBM3 use independent 32-bit sub-channels; DFI 5.0 introduces per-sub-channel signals (`dfi*_p0`, `dfi*_p1`).
- **ALERT_n:** Dedicated DFI signal for DDR5 on-die ECC error reporting.
- **RFM (Refresh Management):** DFI 5.1 adds `dfi_rfm_req/ack` for fine-grained refresh control to mitigate Row Hammer.
- **PPR (Post Package Repair):** DFI 5.1 adds hard/soft PPR control signals.
- **Frequency Set Point (FSP):** LPDDR4/5 multi-rate mode fully supported.

8. Memory Controller — AXI Interface

8.1 AXI Protocol Overview for Memory Controllers

A DDR Memory Controller (DDRC) typically exposes one or more **AXI4 Slave ports** to the SoC interconnect (e.g., ARM NIC-400, CoreLink CCI, CHI interconnect). The AXI slave accepts read/write transactions from CPUs, GPUs, DMA engines, and other bus masters. Internally, the MC translates AXI bursts into DRAM commands (ACT, RD, WR, PRE) and schedules them through its command queue.

8.2 AXI4 Channels & Signal Summary

Channel	Signal	Direction (to MC)	Description
AW (Write Address)	AWID[n:0]	Master→MC	Write transaction ID
	AWADDR[31:0/63:0]	Master→MC	Write start address
	AWLEN[7:0]	Master→MC	Burst length (0=1, 255=256 beats)
	AWSIZE[2:0]	Master→MC	Bytes per beat (0=1, 6=64)
	AWBURST[1:0]	Master→MC	Burst type: FIXED/INCR/WRAP
	AWVALID	Master→MC	Address valid
	AWREADY	MC→Master	MC accepts address
	WDATA[w-1:0]	Master→MC	Write data (32/64/128/256/512b)
	WSTRB[w/8-1:0]	Master→MC	Byte write strobes
	WLAST	Master→MC	Last beat in burst
	WVALID	Master→MC	Write data valid
	WREADY	MC→Master	MC accepts write data
B (Write Response)	BID[n:0]	MC→Master	Response transaction ID
	BRESP[1:0]	MC→Master	OKAY/EXOKAY/SLVERR/DECERR
	BVALID	MC→Master	Response valid
	BREADY	Master→MC	Master accepts response
	ARID[n:0]	Master→MC	Read transaction ID
	ARADDR[31:0/63:0]	Master→MC	Read start address
	ARLEN[7:0]	Master→MC	Burst length
	ARSIZE[2:0]	Master→MC	Bytes per beat
	ARBURST[1:0]	Master→MC	Burst type
	ARVALID	Master→MC	Address valid
	ARREADY	MC→Master	MC accepts address
R (Read Data)	RID[n:0]	MC→Master	Read data ID
	RDATA[w-1:0]	MC→Master	Read data
	RRESP[1:0]	MC→Master	Read response status
	RLAST	MC→Master	Last beat
	RVALID / RREADY	Both	Handshake

Table 13: AXI4 Signals at Memory Controller Slave Port

8.3 AXI Extended Signals (QoS, Security, Cache)

Signal	Direction	Description	AXI Version
AWQOS[3:0] / ARQOS[3:0]	Master→MC	Quality of Service priority (0–15)	AXI4
AWPROT[2:0] / ARPROT[2:0]	Master→MC	Privilege/Security/Data-type bits	AXI3+
AWCACHE[3:0] / ARCACHE[3:0]	Master→MC	Cache attribute: bufferable/modifiable	AXI3+
AWLOCK / ARLOCK	Master→MC	Exclusive access lock bit	AXI3/4
AWREGION[3:0] / ARREGION[3:0]	Master→MC	Address region decode	AXI4
AWUSER[n:0] / ARUSER[n:0]	Master→MC	User-defined sideband signals	AXI4
WUSER[n:0]	Master→MC	Write data user bits	AXI4
RUSER[n:0] / BUSER[n:0]	MC→Master	Read/resp user sideband	AXI4

Table 14: AXI Extended/Sideband Signals

8.4 AXI vs AXI-Lite vs AXI4-Stream at MC

Feature	AXI4	AXI4-Lite	AXI4-Stream	CHI (ARM)
Burst Transfers	✓ Yes	× No	✓ Yes	✓ Yes
Out-of-Order	✓ Yes	× No	N/A	✓ Yes
Exclusive Access	✓ Yes	Limited	× No	✓ Yes
QoS Support	✓ Yes	× No	× No	✓ Yes
Typical MC Use	Primary data port	Reg access/cfg	Streaming data	ARM big.LITTLE SoC
Data Width	32–512b	32/64b	Parameterized	128–512b
ID Width	Parameterized	0	N/A	Parameterized
Ordering Model	ID-based	Single	Stream	Snoop-based

Table 15: AXI Interface Variants Used with Memory Controllers

9. Memory Controller PHY Interface — Complete Signal Reference

9.1 PHY Interface Architecture

The MC-PHY interface is standardised as DFI (Section 5/6). The PHY performs: (1) **Serialisation/De-serialisation** of DFI data to DDR I/O, (2) **Delay-locked loop (DLL)** and **PLL** management, (3) **ODT** and **impedance calibration (ZQ)**, (4) **Training algorithms** (write leveling, gate training, read leveling, VREF), (5) **DQS gating** and **per-bit deskew**, and (6) interfacing DRAM pins directly.

9.2 PHY Interface Signal Categories (DFI 5.x)

Category	MC→PHY Signals	PHY→MC Signals	DFI Ver.
Clocking	dfi_clk, dfi_clk_en	dfi_pll_lock	1.0
Cmd/Address	dfi_address, dfi_bank, dfi_bg, dfi_act_n, dfi_cs_n, dfi_cke, dfi_odt, dfi_ras_n, dfi_cas_n, dfi_we_n	dfi_alert_n	1.0–5.0
Write Data	dfi_wrdata, dfi_wrdata_en, dfi_wrdata_mask, dfi_wrdata_cs_n	–	1.0–5.0
Read Data	dfi_rddata_en, dfi_rddata_cs_n	dfi_rddata, dfi_rddata_valid	1.0–5.0
Write Leveling	dfi_wrlvl_en, dfi_wrlvl_strobe	dfi_wrlvl_resp	2.1
Gate Training	dfi_rdlvl_gate_en	dfi_rdlvl_gate_resp	3.0
Read Leveling	dfi_rdlvl_en, dfi_rdlvl_load, dfi_rdlvl_run, dfi_rdlvl_delay	dfi_rdlvl_resp, dfi_rdlvl_result	3.0
CA Training	dfi_calvl_en, dfi_calvl_capture	dfi_calvl_resp, dfi_calvl_result	3.1
VREF Training	dfi_vref_en, dfi_vref_value, dfi_vref_cs	dfi_vref_resp	4.0
PHY Update	dfi_ctrlupd_req, dfi_ctrlupd_ack	dfi_phyupd_req, dfi_phyupd_ack	3.0
Init	dfi_init_start, dfi_freq_ratio	dfi_init_complete	2.1
Low Power	dfi_lp_ctrl_req, dfi_lp_ctrl_wakeup, dfi_lp_data_req, dfi_lp_data_wakeup	dfi_lp_ctrl_ack, dfi_lp_data_ack	2.0
Freq Change	dfi_freq_fsp, dfi_freq_ratio	–	3.0/4.0
Error / Status	dfi_error_info	dfi_error, dfi_alert_n	3.0
ECC (DDR5)	dfi_wrdata_ecc	dfi_rddata_ecc, dfi_ecc_err	5.0
RFM/PPR	dfi_rfm_req, dfi_ppr_req	dfi_rfm_ack, dfi_ppr_ack	5.1
Sub-channel	dfi*_p0, dfi*_p1 signals	dfi*_p0, dfi*_p1 signals	5.0

Table 16: Complete PHY Interface (DFI 5.x) Signal Categories

9.3 DFI Timing Parameters (MC must configure these)

Parameter	Description	Typical Value (DDR4)
<code>dfi_t_rddata_en</code>	Cycles from RD cmd to <code>dfi_rddata_en</code>	CL - PHY_RD_LAT
<code>dfi_t_ctrl_delay</code>	Delay from cmd on DFI to DRAM pin	2-4 DFI cycles
<code>dfi_t_wrdata_delay</code>	Delay from WR cmd to write data on DFI	WL - 1
<code>dfi_t_dram_clk_enable</code>	Cycles to re-enable DRAM clk	1-4 cycles
<code>dfi_t_dram_clk_disable</code>	Cycles to disable DRAM clk	1-4 cycles
<code>dfi_t_wrlvl_en</code>	Delay before sampling write level resp	PHY specific
<code>dfi_t_wrlvl_ww</code>	Min cycles between WL strobes	≥ 4 cycles
<code>dfi_t_rdlvl_en</code>	Delay before read level response valid	PHY specific
<code>dfi_t_phyupd_resp</code>	MC must respond to PHY update request	≤ 4096 cycles
<code>dfi_t_ctrlupd_interval</code>	Min cycles between ctrl updates	$\geq tRFC$

Table 17: DFI Timing Parameters Configured by Memory Controller

10. Master Reference Table — All DRAM Types, DFI, AXI, PHY

DRAM Type	VDD (V)	Data Rate (MT/s)	Peak BW (GB/s)	Prefetch/B	Min DFI Ver.	Ideal DFI Ver.	AXI Data Width	AXI Addr Width	PHY Training	On-Die ECC	Key Use Case
DDR1	2.5	200–400	3.2	2n/BL2-4	1.0	2.1	64b	32b	Basic	× No	Desktop 2000s
DDR2	1.8	400–1066	8.5	4n/BL4-8	1.0	3.0	64b	32b	WL only	× No	Desktop 2003
DDR3	1.5	800–2133	17	8n/BL8	2.1	3.1	64/128b	32b	WL+RL+	× No	Server/PC
DDR4	1.2	1600–3200	25.6	8n/BL8	3.1	4.0	128/256b	40b	Full+VRE	× No	Server/PC
DDR5	1.1	3200–8400	67.2	16n/BL16	5.0	5.1	256/512b	40b	Full+ECC	✓ Yes	Server/AI
LPDDR1	1.8	200–400	3.2	1n/BL2-8	2.0	2.1	32b	32b	Basic	× No	Mobile
LPDDR2	1.2	500–1066	8.5	4n/BL4-16	2.1	3.0	32/64b	32b	WL+RL	× No	Mobile
LPDDR3	1.25	800–2133	17	8n/BL8	3.1	4.0	64b	36b	WL+RL+	× No	Mobile
LPDDR4	1.1	1600–4267	34	16n/BL16	4.0	5.0	64/128b	40b	Full+FSP	× No	Mobile
LPDDR4X	0.6	2133–4267	34	16n/BL16	4.0	5.0	64/128b	40b	Full+FSP	× No	Mobile
LPDDR5	0.5	3200–6400	51.2	16n/BL16-32	5.0	5.1	128/256b	40b	Full+LECC	✓ Yes	Mobile/AI
LPDDR5X	0.5	6400–8533	68.3	16n/BL16	5.1	5.1	128/256b	40b	Full+LECC	✓ Yes	Mobile AI
GDDR5	1.5	5000–8000	64	8n	4.0	4.0	256b	40b	WL+RL	× No	GPU
GDDR6	1.25	12000–16000	128	16n	5.0	5.1	256/512b	40b	Full	× No	GPU/AI
HBM1	1.2	1000	128	–	3.1	4.0	512b	40b	Limited	✓ Yes	HPC/GPU
HBM2	1.2	2000	256	–	4.0	5.0	512b	40b	Full	✓ Yes	AI/HPC
HBM2E	1.2	3600	460	–	5.0	5.0	512b	40b	Full	✓ Yes	AI/HPC
HBM3	1.1	6400	819	–	5.0	5.1	512/1024b	40b+	Full+sub-ch	✓ Yes	AI/Data Ctr

Table 18: Master Reference: All DRAM Types with DFI, AXI, and PHY Parameters

11. PHY Training Algorithms — By DDR Generation

Training Algorithm	DDR3	DDR4	DDR5	LPDDR	LPDDR	LPDDR	HBM2/3
Write Leveling	✓ Yes	✓ Yes	✓ Yes	✓ Yes	✓ Yes	✓ Yes	✓ Yes
Gate Training	✓ Yes	✓ Yes	✓ Yes	✓ Yes	✓ Yes	✓ Yes	✓ Yes
Read Leveling	✓ Yes	✓ Yes	✓ Yes	✓ Yes	✓ Yes	✓ Yes	✓ Yes
Write DQ Deskew	✓ Yes	✓ Yes	✓ Yes	✓ Yes	✓ Yes	✓ Yes	✓ Yes
Read DQ Deskew	✓ Yes	✓ Yes	✓ Yes	✓ Yes	✓ Yes	✓ Yes	✓ Yes
CA Training	× No	✓ Yes	✓ Yes	✓ Yes	✓ Yes	✓ Yes	N/A
VREF (DRAM side)	× No	✓ Yes(MF)	✓ Yes	× No	✓ Yes	✓ Yes	✓ Yes
VREF (PHY side)	× No	✓ Yes	✓ Yes	× No	✓ Yes	✓ Yes	✓ Yes
DFE (Decision FB)	× No	Optional	✓ Yes	× No	× No	Optional	✓ Yes
Link ECC Training	× No	× No	× No	× No	× No	✓ Yes	✓ Yes
Per-bit RX Margin	✓ Yes	✓ Yes	✓ Yes	✓ Yes	✓ Yes	✓ Yes	✓ Yes
2D Eye Scan	Optional	Optional	✓ Yes	Optional	Optional	✓ Yes	✓ Yes

Table 19: PHY Training Requirements by DRAM Generation

12. System Architecture: CPU → AXI → MC → DFI → PHY → DRAM

```

+-----+
| SoC Interconnect (ARM NIC / CHI / APB) |
| | |
| v AXI4 (AWADDR, WDATA, RDATA, ARADDR ...) |
| +-----+ |
| | DDR Memory Controller (DDRC) | | | | | |
| | +-----+ +-----+ | |
| | | AXI Slave IF | | Command Scheduler | | |
| | | (AR/AW/W/B/R) | | (ACT, RD, WR, PRE, REF) | | |
| | +-----+ +-----+ | |
| | +-----+ | |
| | | DFI 5.x Interface (MC -> PHY) | | |
| | | dfi.address, dfi.wdata, dfi.lp_ctrl_req | | |
| | +-----+ | |
| +-----+ |
| | DFI Bus |
| v |
| +-----+ |
| | DDR PHY (Physical Layer) | | | | | | | |
| | +-----+ +-----+ +-----+ | |
| | | PLL / DLL | | TX Drv | | DQS Capture / | | |
| | | Clk Gen | | (CK,DQ) | | Deskew Logic | | |
| | +-----+ +-----+ +-----+ | |
| | +-----+ | |
| | | ZQ Cal / ODT / VREF generation | | |
| | +-----+ | |
| +-----+ |
| | DRAM Pins: CK, CKE, CS, RAS, CAS, WE, DQ, DQS, ODT |
| v |
| +-----+ |
| | DRAM Device (DDRx / LPDDRx / HBM) | |
| +-----+ |
+-----+

```

12.1 MC Internal Architecture (AXI to DFI)

MC Block	Function	Key Interfaces
AXI Slave	Accepts burst R/W from bus mas- ters; buffers in RD/WR queues	AXI4: 5-channel handshake
Address Map	Translates AXI addresses to rank/bank/row/col	AXI ARADDR/AWADDR
Command Scheduler	Reorders DRAM commands for max efficiency (FR-FCFS, policy-based)	Internal queues
Refresh Engine	Generates REF commands per tREFI, handles RFM/pREF	DFI refresh signals
Training FSM	Orchestrates PHY training se- quences at init and runtime	DFI training signals
ECC Engine	SECDED (DDR4 RDIMM) or link ECC for DDR5/LPDDR5	Internal + DFI ECC
Power Mgmt	CKE control, self-refresh, power- down, frequency change	DFI LP interface
DFI Adapter	Converts MC internal timing to DFI cycle-accurate signals	DFI all channels

Table 20: Memory Controller Internal Functional Blocks

13. Quick Reference Summary

13.1 DFI Version — Feature Flags Quick-Ref

Feature	2.1	3.0/3.1	4.0	5.0	5.1
Write Leveling (dfi_wrlvl_*)	✓ Yes	✓ Yes	✓ Yes	✓ Yes	✓ Yes
Gate Training (dfi_rdlvl_gate_*)	× No	✓ Yes	✓ Yes	✓ Yes	✓ Yes
Read Leveling (dfi_rdlvl_*)	✓ Yes	✓ Yes	✓ Yes	✓ Yes	✓ Yes
CA Leveling (dfi_calvl_*)	× No	✓ Yes	✓ Yes	✓ Yes	✓ Yes
VREF Training (dfi_vref_*)	× No	× No	✓ Yes	✓ Yes	✓ Yes
Freq Change (dfi_freq_*)	× No	✓ Yes	✓ Yes	✓ Yes	✓ Yes
Sub-channel (dfi*_p[01])	× No	× No	× No	✓ Yes	✓ Yes
DRAM ECC (dfi_alert_n)	× No	× No	× No	✓ Yes	✓ Yes
RFM (dfi_rfm_*)	× No	× No	× No	× No	✓ Yes
PPR (dfi_ppr_*)	× No	× No	× No	× No	✓ Yes
LP Interface (dfi_lp_*)	✓ Yes	✓ Yes	✓ Yes	✓ Yes	✓ Yes
PHY/Ctrl Update (dfi*_upd_*)	× No	✓ Yes	✓ Yes	✓ Yes	✓ Yes
Init Interface (dfi_init_*)	✓ Yes	✓ Yes	✓ Yes	✓ Yes	✓ Yes

Table 21: DFI Feature Flags — Quick Reference by Version

13.2 AXI MC Interface Key Configuration Points

- **AXI data width** should match DDR bus width $\times$ prefetch factor (e.g., DDR4: 64b bus, 8n prefetch $\Rightarrow$ 512b AXI data beats handled via burst).
- **AWLEN/ARLEN**: Set burst length so one AXI burst fills exactly one or multiple DRAM burst (avoid crossing page boundaries).
- **AWBURST=INCR** is standard; WRAP is used for cache-line fills; FIXED for FIFO-style peripherals.
- **QoS**: Use AWQOS/ARQOS to prioritise latency-sensitive traffic (display, ISP) over throughput traffic (CPU speculative).
- **Exclusive access**: ARM CPUs use ARLOCK/AWLOCK for LL/SC primitives; MC must implement exclusive monitor.
- **ECC**: DDR5 on-die ECC transparent; DDRC ECC requires wider AXI or internal ECC scrub logic.

14. Advanced Topics

14.1 Row Hammer Mitigation in Modern DRAM (DFI 5.1 RFM)

Row Hammer is a DRAM vulnerability where repeated activation of a row induces bit flips in adjacent rows. DFI 5.1 introduces **Refresh Management (RFM)**: the MC tracks activation counts per bank and issues RFM commands when a threshold is reached.

Mechanism	DRAM Standard	DFI Signals
pTRR (targeted refresh)	DDR4/5 (JEDEC)	Via normal REF commands, MC manages
RFM (Refresh Mgmt)	DDR5, LPDDR5	<code>dfi_rfm_req</code> , <code>dfi_rfm_ack</code> , <code>dfi_rfm_cs_n</code>
ALERT_n (ECC error)	DDR5	<code>dfi_alert_n</code> (PHY→MC)
DRFM (directed)	LPDDR5, DDR5	Encoded via <code>dfi_rfm_*</code>

Table 22: Row Hammer Mitigation and DFI 5.1

14.2 LPDDR5 vs DDR5 — DFI Interface Differences

Aspect	DDR5 (DFI 5.0+)	LPDDR5 (DFI 5.0+)
Sub-channel	2×32b via <code>dfi_*_p[01]</code>	2×16b via <code>dfi_*_p[01]</code>
CA Bus	<code>dfi_address[13:0]</code>	<code>dfi_address[5:0]</code> (6-bit CA)
Freq Set Point	Not used	<code>dfi_freq_fsp[1:0]</code>
Link ECC	Separate channel	<code>dfi_wrdata_ecc</code> / <code>rddata_ecc</code>
Power Down	Standard LPDDR	CKE-gated + deepsleep modes
Training seq	CA→WL→GT→RL→VREF	WL→CA→GT→RL→VREF
DFI.RESET_n	✓Yes	✓Yes

Table 23: DDR5 vs LPDDR5 DFI Implementation Differences

14.3 HBM3 Specific PHY Interface Notes

- HBM3 uses **pseudo channels**: each 64b channel is split into two 32b pseudo-channels, requiring per-pseudo-channel DFI signals (`dfi*_p[01]`).
- HBM3 PHY uses **Through-Silicon Vias (TSVs)** — no traditional package-level DQ routing; PHY is integrated in the base logic die.
- HBM3 supports **DBI (Data Bus Inversion)**, **on-die ECC (SECDED)**, and **data scrambling** — all signalled via DFI 5.x.
- **Temperature monitoring**: HBM3 devices include on-die thermal sensors; read via MRS registers through DFI command channel.
- **AXI interface**: HBM3 MC uses extremely wide AXI or proprietary interface (512b–1024b) with very high outstanding transaction depths.

15. References

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- [4] DFI PHY Interface Working Group, *DFI 5.1 Specification*, 2022.
- [5] DFI PHY Interface Working Group, *DFI 4.0 Specification*, 2016.
- [6] ARM Ltd., *AMBA AXI and ACE Protocol Specification AXI3, AXI4, and AXI4-Lite*, ARM IHI0022H, 2021.
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*Prepared for VLSI Testing & Testability — PES University, ECE Department
March 2026 · All specifications subject to JEDEC/DFI working group updates*